

Classical Complex Systems, Final Project:

Project 1: Lane formation in binary mixture driven by an external field

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Outline:

Introduction/Motivation

Theoretical background

Particular questions/goals

Implementation; code and methods

Primary results

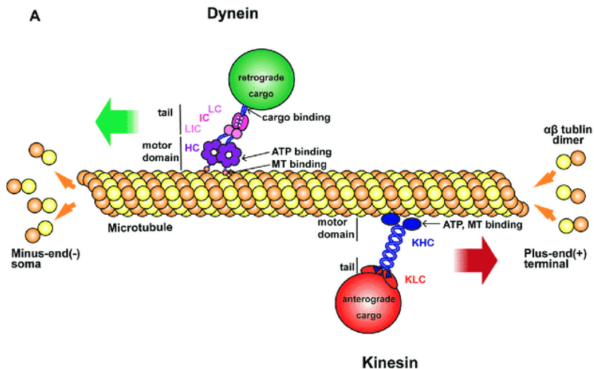
Playing around

Discussion and Summary

Introduction and Motivation



Introduction and Motivation



Theoretical background

Standard Brownian (overdamped) dynamics setup in 2D, plus pairwise Yukawa and const. driving:

$$\gamma \frac{d\mathbf{r}_i}{dt} = - \sum_{j \neq i} \nabla_{\mathbf{r}_i} V(r_{ij}) + \mathbf{F}_i^{\text{DC}} + \mathbf{F}_i^{(\text{R})}(t) \quad (1)$$

$$\begin{aligned} \left\langle \frac{1}{\gamma} F_{i\alpha}^{(\text{R})}(t) \frac{1}{\gamma} F_{j\beta}^{(\text{R})}(t') \right\rangle &= 2 \frac{k_B T}{\gamma} \delta_{ij} \delta_{\alpha\beta} \delta(t - t') \\ &= 2D \delta_{ij} \delta_{\alpha\beta} \delta(t - t') \end{aligned} \quad (2)$$

$$V(r) = V_0 \frac{\sigma}{r} e^{-\kappa(r-\sigma)} \quad (3)$$

$$\mathbf{F}_i^{\text{DC}} = f_i F_D \hat{\mathbf{x}}, \quad f_i \in \{-1, +1\} \quad (4)$$

Update rule:

$$\mathbf{r}_i(t + \Delta t) = \mathbf{r}_i(t) + \sqrt{2D\Delta t}\mathbf{N}_i(t) + \frac{1}{\gamma}\mathbf{F}_i^{\text{DC}} + \frac{1}{\gamma}\mathbf{F}_i^{\text{Yuk}} \quad (5)$$

$$\mathbf{F}_i^{\text{Yuk}} = \sum_{j \neq i} V_0 \frac{\sigma}{r_{ij}^2} e^{-\kappa(r_{ij} - \sigma)} (1 + \kappa r_{ij}) \frac{\mathbf{r}_i - \mathbf{r}_j}{r_{ij}}, \quad r_{ij} = |\mathbf{r}_i - \mathbf{r}_j| \quad (6)$$

NB: Velocity associated with pure BD;

$$\begin{aligned} \mathbf{v}_{\text{BD}}(t + \frac{1}{2}\Delta t) &= \frac{\mathbf{r}_i(t + \Delta t) - \mathbf{r}_i(t)}{\Delta t} \\ &= \sqrt{\frac{2D}{\Delta t}} \mathbf{N}, \end{aligned} \quad (7)$$

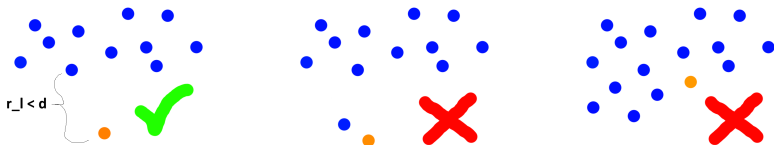
so $v_{\text{BD}} \sim \sqrt{2D/\Delta t}$; need strong driving!

(Note: $v_{\text{BD}} = \sqrt{\frac{2}{\pi}} \sqrt{\frac{2D}{\Delta t}}$.)

Order parameter; characterize onset of lane formation

$$\phi_i = \begin{cases} 1, & \text{if } |y_i - y_j| > r_\ell \quad \forall j \text{ with } f_j \neq f_i, \\ 0, & \text{otherwise,} \end{cases} \quad (8)$$

$$\Phi = \sum_{i=1}^N \phi_i, \quad r_l = \frac{1}{2\sqrt{\rho}}. \quad (9)$$



Questions:

- ▶ $\Phi \leftrightarrow \langle v \rangle$ correspondence?
- ▶ $\Phi \leftrightarrow F_D$ correspondence?

Questions/goals

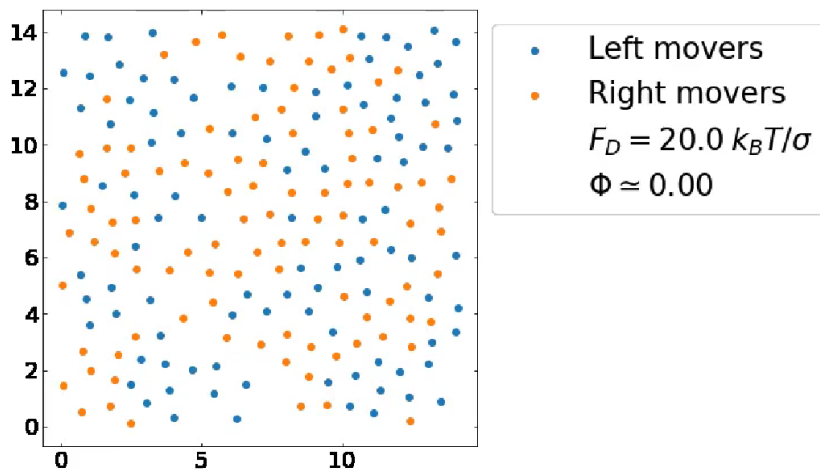
- ▶ Is the proposed order parameter able to characterize the onset of lane formation?
- ▶ How is the relationship between the order parameter and external driving force?
- ▶ How is the relationship between the external driving force and the mean velocities?
- ▶ How is the relationship between the order parameter and screening range?

Implementation

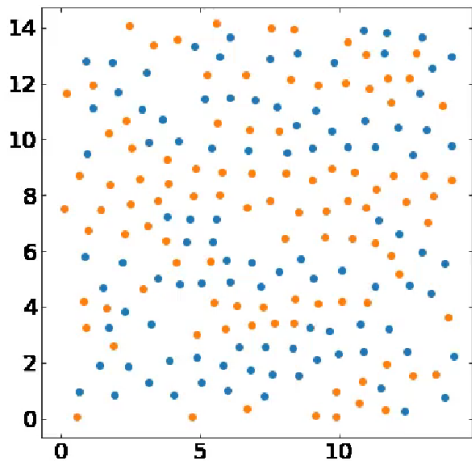
- ▶ How to differentiate two groups with two different direction?
+1 and -1 species
- ▶ How to implement external driving force?
Multiply the external driving force $\mathbf{F}_i^{\text{DC}}$ with the species value associated with the particle
- ▶ Update algorithm
 - 1: **for** *particle* = 1... *N* **do**
 - 2: $\text{pos}[\text{particle}] \leftarrow$ Particle Position
 - 3: $r \leftarrow$ Distance (minimum image convention)
 - 4: $\mathbf{F}_i^{\text{Yuk}} \leftarrow$ Yukawa Force
 - 5: $\text{pos}[\text{particle} + 1] \leftarrow \text{pos}[\text{particle}] + \Delta t(\mathbf{F}_i^{\text{Yuk}} + \mathbf{F}_i^{\text{DC}} + \mathbf{F}_i^{\text{R}})$
 - 6: **end for**

Analysis results: Phases

Stuck / Jammed:



'Moving liquid':



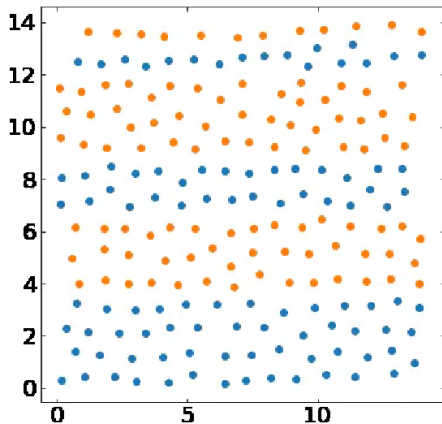
• Left movers

• Right movers

$$F_D = 60.0 k_B T / \sigma$$

$$\Phi \approx 0.18$$

Well-formed lanes::



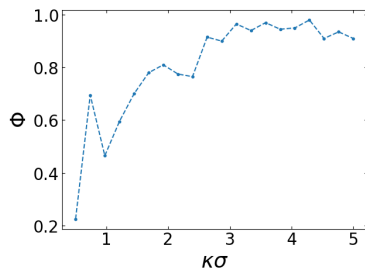
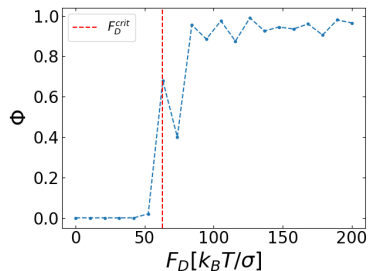
Left movers

Right movers

$$F_D = 100.0 k_B T / \sigma$$

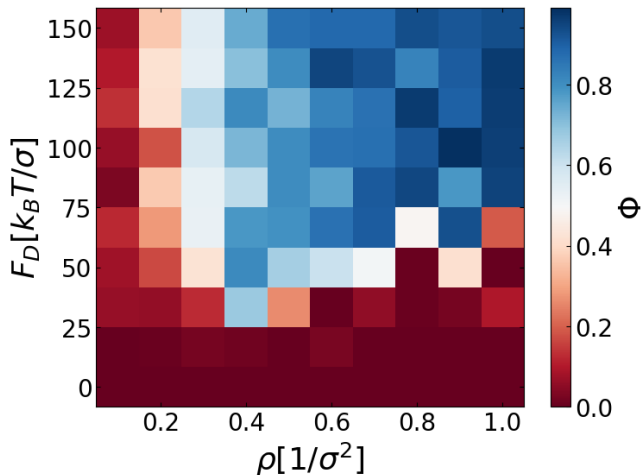
$$\Phi \approx 0.90$$

Analysis results: Lane Formation Φ



- ▶ Phase transition with $F_D^{crit} \approx 63 k_B T / \sigma$
- ▶ $V(r) = V_0 \frac{\sigma}{r} e^{-\kappa(r-\sigma)}$, $F(r) = V_0 \frac{\sigma}{r} e^{-\kappa(r-\sigma)} (\kappa + \frac{1}{r})$

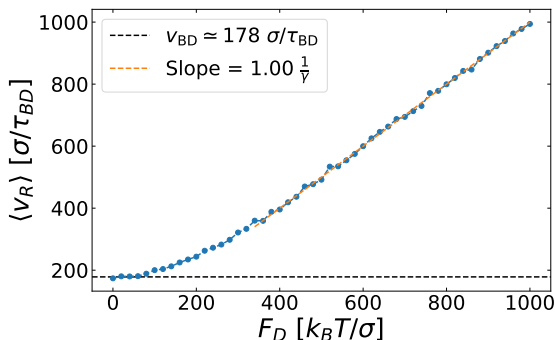
Analysis results: Phase Diagram



- ρ dependence. Low $\rho \rightarrow$ lot of spaces, Large $\rho \rightarrow$ lot of collisions

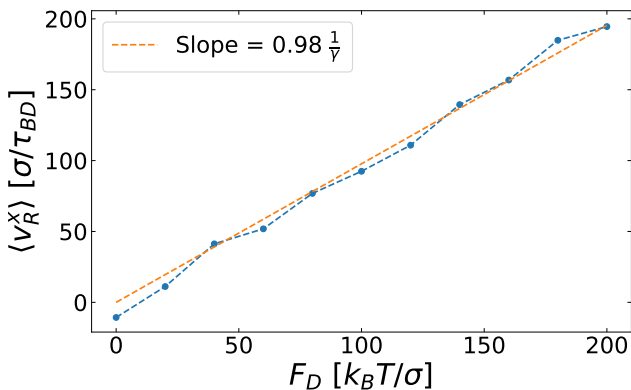
Analysis results: Velocities

Naïve expectation: $\langle v \rangle \simeq \frac{1}{\gamma} F_D$

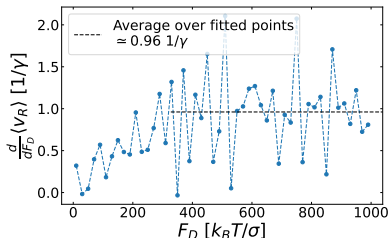


→ BD speed overpowers for small driving, asymptotic limit is the 'linear response' (overdamped)

However, for *signed* velocity;

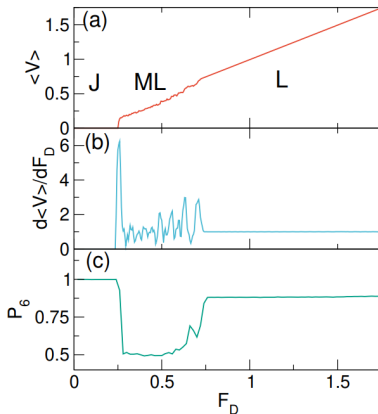


Comparison with literature: Look for signs of liquid behaviour in the velocity derivative



(a) Derivative plot of speed (unsigned) per driving force

$$V(r) = V_0 \frac{\sigma}{r} e^{-\kappa(r-\sigma)}$$



(b) Velocity derivative reported in literature (Reichhardt & Reichhardt) $V(r) = 1/r - Be^{-\kappa r}$ (SALR)

Extra: Central obstruction

'Particle' fixed at $(x, y) = (L/2, L/2)$ with $\mathbf{F}_{\text{circ}}(r) = 50\mathbf{F}_{\text{Yuk}}(r)$ on other particles.

Questions:

- ▶ Order parameter?
- ▶ Curved lanes?
- ▶ Other jammed phases?

→ Animations!

NB: $\rho = 3/\sigma^2$, $F_D = 100 \frac{k_B T}{\gamma}$, κ varied.

Possible Improvements

- ▶ **Computation power**
- ▶ Smoothing of order parameter
- ▶ Langevin (inertia)
- ▶ Variance/autocorrelation smoothing of noise

Summary

- ▶ The proposed order parameter Φ can be used to quantify lane formation when there are no obstructions.
- ▶ The phase transition *cannot* be characterized based on a simple finite-difference measured speed, but the x-velocity is one-to-one with the driving force
- ▶ Phase transition (discontinuity in Φ) at $F_D^{\text{crit}} \rightarrow \rho$ -dependent
- ▶ Screening range affects lane formation
- ▶ Documentation:
<https://github.com/prakhosha/Lane-Formation>

References

- ▶ Laurent Guillaud, Sara Emad El-Agamy, Miki Otsuki, Marco Terenzio; Anterograde Axonal Transport in Neuronal Homeostasis and Disease. 2020.
- ▶ Robin Jose and Ludger Santen; Self-organized lane-formation in bidirectional transport of molecular motors. 2020.
- ▶ J. Dzubiella, G. P. Hoffmann, and H. Lowen; Lane formation in colloidal mixtures driven by an external field. 2001.
- ▶ C. Reichhardt and C. J. O. Reichhardt; Laning Transitions in Pattern Forming Driven Binary Systems with Competing Interactions. 2025.